

Reflection Report: Employee Attrition Analysis

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Batch: D

1. Introduction and What I Learned

This Week 1 project on Employee Attrition Analysis has been an excellent introduction to the real-world application of data science in Human Resources. I learned that data science is not just about writing code; it is fundamentally about solving business problems. By investigating the IBM HR dataset, I learned how to translate vague business concerns—such as "why are employees leaving?"—into specific, data-driven questions. I also learned the critical importance of Exploratory Data Analysis (EDA) as a foundational step before jumping into complex predictive machine learning models. Understanding the "shape" and story of the data is crucial for making accurate business recommendations.

2. Skills Gained

Throughout this assignment, I strengthened several core technical and analytical skills:

Data Manipulation: I gained practical experience using Python's pandas library to load, inspect, and clean datasets, including checking for missing values, duplicates, and understanding data types.

Data Visualisation: I improved my ability to use matplotlib and seaborn to create meaningful visualisations. I learned how to select the right chart for the right data (e.g., using boxplots to compare distributions of income between leavers and stayers, and bar charts for categorical comparisons like departments).

Business Communication: I developed the skill of translating raw statistical outputs and charts into plain-English business insights. For example, rather than just stating, "the median income of leavers is lower," I learned to frame it as a business risk: "lower-paid employees are more likely to resign, indicating a need to review compensation bands."

3. Challenges Faced

One of the main challenges I faced was deciding which variables to plot against each other to extract the most meaningful insights. With 35 columns in the dataset, it was tempting to create charts for every single variable, which would have led to information overload. I had to practice feature selection by focusing on the variables that directly answered the core business questions (such as Age, Monthly-Income, Over-Time, and Department). Another minor challenge was formatting the

visualisations for readability, such as ensuring that long categorical labels (like Job Roles) did not overlap on the x-axis, which I solved by adjusting the chart layouts and orientations.

4. What I Would Improve and Next Steps

If I were to expand on this analysis in the future, I would incorporate statistical testing (such as T-tests or Chi-Square tests) to mathematically validate the visual patterns I observed, ensuring that the differences in attrition rates are statistically significant and not just due to random chance.

Additionally, since the dataset is imbalanced (only ~16% of employees left), I would research and apply techniques like SMOTE (Synthetic Minority Over-sampling Technique) to prepare the data for future predictive modelling. Finally, I would love to connect this Python analysis to a BI tool like Power BI to create an interactive dashboard for the HR stakeholders.

Conclusion

Overall, this first week has been highly rewarding. It has bridged the gap between theoretical data science concepts and practical business consulting, setting a strong foundation for the predictive modelling tasks ahead in this internship.